



A review of failure modes, condition monitoring and fault diagnosis methods for large-scale wind turbine bearings

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ABSTRACT

Large-scale wind turbine bearings including main bearings, gearbox bearings, generator bearings, blade bearings and yaw bearings, are critical components for wind turbines to convert kinetic wind energy into electrical energy. Unlike general-purpose industrial bearings, the loads and rotation speeds of wind turbine bearings change considerably because of dynamic wind flows. In the case of some extreme operating conditions, large-scale wind turbine bearings suffer excessive loads and they can be potentially damaged. Therefore, it is essential to develop reliable and cost-effective condition monitoring and fault diagnosis methods to assess the damage level and failure modes so that a proper maintenance plan can be designed. This paper aims at systematically and comprehensively summarizing current large-scale wind turbine bearing failure modes and condition monitoring and fault diagnosis achievements. Firstly, the representative failure modes of large-scale wind turbine bearings are reviewed in detail which can help to understand the causes and effects of the tribological issues of these bearings. Then, condition monitoring and fault diagnosis methods of large-scale wind turbine bearings are presented; within which failure modes, experimental scale and signal processing approaches are summarized. Finally, a number of popular condition monitoring and fault diagnosis approaches that can be potentially used for wind turbine bearings are reviewed, followed by a brief summary of future research directions for wind turbine bearing fault diagnosis.

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1. Introduction

Wind power, as a reliable and emission-free energy source available in nature, has become the world's fastest-growing renewable energy source leading a way towards a clean energy future [1]. Based on the global wind statistics, the global cumulative installed wind capacity was only 23.9 GW in 2001, but had increased to 486.8 GW by the end of 2016 and it is estimated to be over 800 GW by 2021 [2].

Wind turbines, as shown in Fig. 1, are made up of mechanical and electrical components to extract kinetic wind energy from the available wind flow in the atmosphere and convert it to electric energy. The most widely used wind turbine design is the gearbox-operated wind turbine, as shown in Fig. 1(a). The three blades rotate the main shaft which is connected to the generator via a gearbox. The gearbox can increase rotation speeds up to a typical 1500–1800 rpm, which allows the generator to connect to the electric grid [3]. Another popular design is the direct-drive wind turbine, as shown in Fig. 1(b), which lacks a gearbox. Compared with gearbox operated wind turbines, direct-drive turbines eliminate the most complicated gearbox, but the size of the generator becomes larger [4].

At present, there are over a quarter of a million wind turbines in operation and this number will continue to increase in the future [1]. Additionally, the size of wind turbines will also become larger every year posing remarkable challenges in the operation and maintenance (O&M) of all the components. In particular, the large-scale bearings in wind turbines are critical components supporting heavy loads and handling large moments. Different from general-purpose industrial bearings, these large-scale wind turbine bearings are operated under harsh conditions and mounted at high altitudes with higher failure rates and repair costs [7]. In order to increase wind turbine production and reduce O&M costs, condition monitoring and fault diagnosis (CMFD) of these bearings is vitally important to guarantee reliability in wind turbine operations. However, wind turbines have different types of large-scale bearings in terms of their rotation speeds: (i) very slow-speed bearings (yaw bearings and pitch bearings), (ii) slow-speed bearings (main bearings and gearbox low-speed bearings) and (iii) high-speed bearings (gearbox high-speed bearings, gearbox intermediate-speed bearings and generator bearings). Their different rotation speeds and unique working mechanisms lead to different failure modes. Whereas, as the research on wind turbines is still in an initial stage due to its short history, majority CMFD techniques of

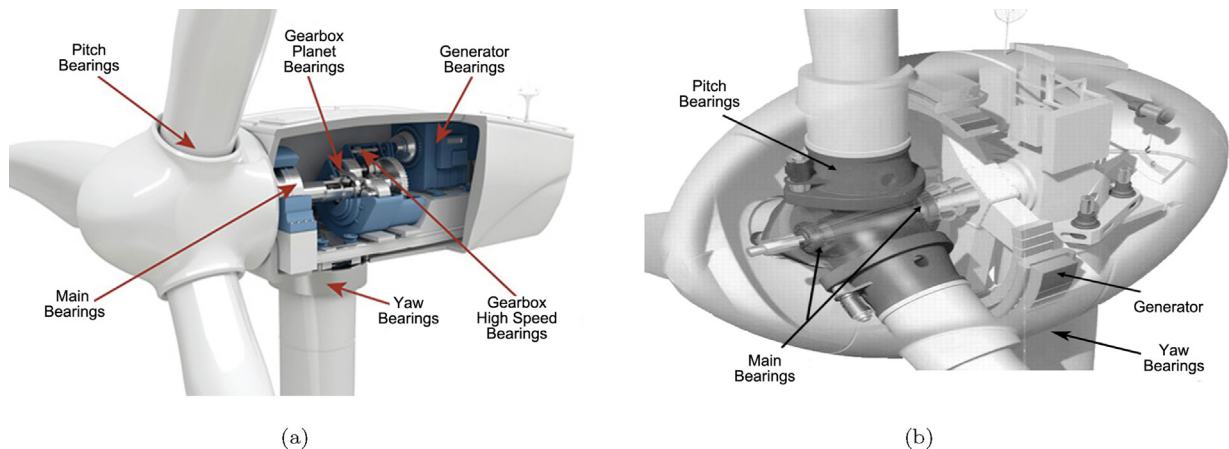


Fig. 1. Wind turbine configuration (a) gearbox operated wind turbine [5] and (b) direct-drive wind turbine [6].

wind turbine bearings mainly focus on indicating the defective components of the bearing (inner race, outer race and balls) without incorporating the failure mode assessment. Thus, it needs more works to enhance the maturity of the developed CMFD techniques. Usually, existing techniques can be easily applied to idealized laboratory environments properly, but may not be utilized for *in situ* CMFD of wind turbine bearings because of complex failure modes and unique bearing working mechanisms. As a result, it is highly important to classify their failure modes, and then develop sophisticated CMFD techniques.

Nonetheless, there is only one review paper available for wind turbine bearing condition monitoring [8], which focuses on reviewing current monitoring techniques and does not review wind turbine bearing failure modes. In terms of our work, we aim at providing a review in two portions.

- The representative failure modes of large-scale wind turbine bearings supported by the real cases and schemata are reviewed. These failure modes are classified which are helpful in assisting readers to understand the causes and effects of bearing faults.
- The existing CMFD methods for five types large-scale bearings, namely main bearings, gearbox bearings, generator bearings, blade bearings and yaw bearings are presented where data acquisition techniques, signal processing methods, main problems, experimental scales and failure modes are carefully reviewed. Furthermore, some state-of-the-art CMFD approaches applied to traditional rotation machines are summarized to make up the shortfall of current CMFD methods for large-scale wind turbine bearings.

The remainder of the review paper is organized in the following way. In Section 2, the types and working conditions of large-scale wind turbine bearings are presented. Section 3 summarizes failure modes and presents visible faults on large-scale wind turbine bearing surfaces. The failure mode classifications of different wind turbine bearings are also concluded in this section. Section 4 gives a detailed review of existing and potential CMFD methods for large-scale wind turbine bearings. Finally, concluding remarks are given in Section 5.

2. Types of large-scale wind turbine bearings

In general, large-scale bearings in wind turbines can be classified into two categories: transmission system bearings and adjustment system bearings. The transmission system bearings including main bearings, gearbox bearings and generator bearings often use roller bearings with a large capacity to carry heavy radial loads. The adjustment system bearings such as blade and yaw bearings have to use slewing bearings to handle both large thrust and radial loads. The details of large-scale wind turbine bearings are introduced below:

- Main bearings:** The main bearings or main shaft bearings are used to hold the hub and the blades; and they can transmit torque to the gearbox. For modern MW-turbines, the size of the main bearings is usually greater than 1 m, which is able to support a heavy radial load; therefore, the types of main bearings are normally chosen as spherical roller bearings, cylindrical roller bearings and tapered roller bearings. These bearings can work in a harsh and unstable environment [9]. To reduce the failure rates, the main bearings are supported through frequent greasing with an adequate amount of lubricant.
- Gearbox bearings:** Wind turbine gearbox bearings are used to support shafts in different gear stages i.e. a low-speed stage,

intermediate-speed stage and high-speed stage. For different gear stages, the choices of bearings are different, which depend on the type of loads. For low-speed stage bearings, they are usually the largest bearings in the gearbox where the loads they support are subject to the wind turbine main bearings. The suitable types of low-speed stage bearings are cylindrical roller bearings and tapered roller bearings. For the intermediate-speed and high-speed stage bearings, they support heavy radial and axial loads. Tapered roller bearings, cylindrical roller bearings, four-point contact bearing and tube roller bearings are suitable for this application [10].

- Generator bearings:** The generator bearings support the shaft in the generator. They are operated at a high speed, but should have low vibration and good insulating properties [11–13]. Electrical pitting is the first concern when designing generator bearings [14]. In order to solve this issue, there are two common solutions for installing electrically insulated bearings. One is a coated bearing which is designed to perform an insulated treatment on the bearing's ring surface. Ceramic and conductive microfibers are two frequently used insulation materials mounted on the bearing's ring surface. The other solution involves hybrid bearings, where the rolling elements are made by ceramic balls, but the rest of the unit is metal. In addition, some tactics, such as using a conductive lubricant and grounding the bearings, can also reduce electrical damage to generator bearings [15].

- Blade bearings/pitch bearings:** The functions of blade bearings are to pitch the blades, optimize the electrical energy output, and stop the wind turbine for protection if the wind speeds are larger than a cut-out speed. In modern wind turbines, blade bearings are either single line or double lines four-point contact ball slewing bearings with either internal gear teeth arrangements (electrical pitch systems) or no teeth arrangements (hydraulic pitch systems) [16,17]. The outer bearing ring is fixed to the hub and the blade is mounted on the inner bearing ring [18].

- Yaw bearings:** The yaw bearings in wind turbines are used to angularly realign the nacelle into the predominant wind direction in order to optimize power input [19,20]. The types of yaw bearings are a single line or double lines four-point contact ball slewing bearings with external gear teeth arrangements. Electric motors drive a yaw bearing via the gear system to enable wind turbines to track and follow the wind direction.

3. Failure modes of large-scale wind turbine bearings

For large-scale wind turbine bearings, different-types of bearings may have different failure modes due to their different designed mechanism and operation conditions. Furthermore, each type of bearings could have multiple failure modes. For these two reasons, it is highly desirable to clearly classify the failure modes in terms of failure mode characteristics and to understand how different data acquisition techniques and signal processing methods can cope with these failure modes, so as to effectively diagnose the faulty bearings of wind turbines. As a result, the investigations of representative failure modes of large-scale wind turbine bearings are presented.

3.1. Plastic deformation

Plastic deformation is very complicated and its progress is irreversible [21]. It comes from the motion of dislocations and is controlled by a thermally-activated motion from a large segment of the stress-temperature region [21,22]. For plastic deformations of bearings, there are two different types. The first is on a macroscale,

and happens over a large area of the contact footprint because of either heavy load or misalignment. The second type is on a micro-scale where plastic deformation only occurs over a small part of the contact footprint, such as indentation [23]. The specific types of plastic deformation are explained as follows:

- a) **Excessive load/overloading:** Excessive load or overloading can cause some areas of the bearing to wear faster than others [24,25]. For large-scale wind turbine bearings, transient wind loads can easily give rise to overloading, thus leading to plastic deformations in a bearing. The position of the deformation due to overloading relates to the pitch of the rolling elements [23]. Fig. 2 displays the damaging consequence of excessive loads on wind turbine gearbox bearing rolling elements.
- b) **Misalignment:** Misalignment happens when raceway edges are not parallel to the ball wear path which can lead to overheating, vibration and deformation [26]. One common reason for the misalignment is that the crankshaft has elastic deflection which is subjected to inertial force [27]. Fig. 3 shows a schema of the misalignment which indicates that the ball bath runs from one side of the raceway to the other. Misalignment is a very common problem for transmission system bearings as the bearings are mounted on rubber bushings. The bushings may be deflected under the condition of large torque and extreme environment, both of which will cause bearing misalignment [28].
- c) **Loose fit/incorrect seating fit:** Loose fit occurs if a bearing ring is in relative motion to a mounting shaft or house. If the relative motion continues vibrating, fretting will happen and can further lead to adhesive wear and overheating [23,29,30]. If the issue is not solved in a timely manner, the looseness will become worse, leading to noise and cracking [30]. Loose fit also happens on transmission system bearings because of the deterioration of rubber bushings.



Fig. 2. The damaging consequence of excessive load on gearbox bearings roller elements [31].

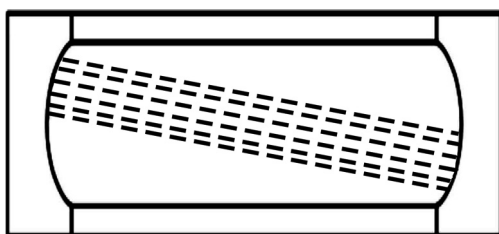


Fig. 3. An example of misalignment.

3.2. Lubricant

The purpose of the lubricant is to protect the rolling contact surface with a piece of thin oil film, avoiding balls contacting the raceway directly. An effective lubricant can reduce wears, transport heat generated by friction, extend service life and keep contamination away from bearings [32]. For these purposes, the lubricant is key to maintaining wind turbine bearings, helping them to continue working properly. The followings are two common failure modes:

- a) **Lubricant failure:** The causes of lubricant failures include insufficient lubricant, over-lubricate, inadequate lubricate and using the wrong types of lubricant [33]. Different lubricants may be incompatible even though each solo lubricant is completely adequate [33]. The direct symptom of lubricant failure is that the surface of ball tracks or balls change to dark color with the odor of burnt petroleum oil [24,33]. Finally, lubricant failures can lead to excessive wear, overheating, noise and the

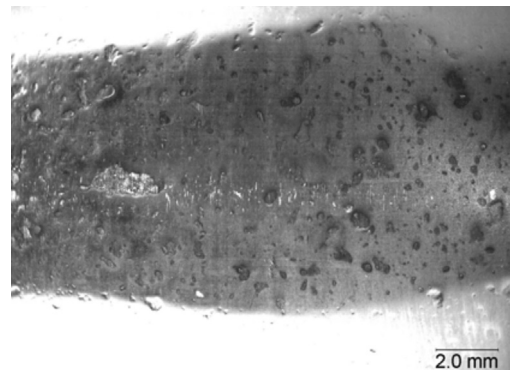


Fig. 4. Embedded wear debris particles on the wind turbine gearbox bearing roller surface [31].

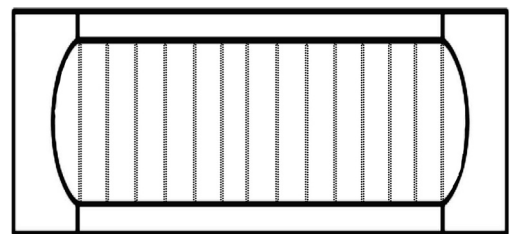


Fig. 5. A schema of electrical fluting.

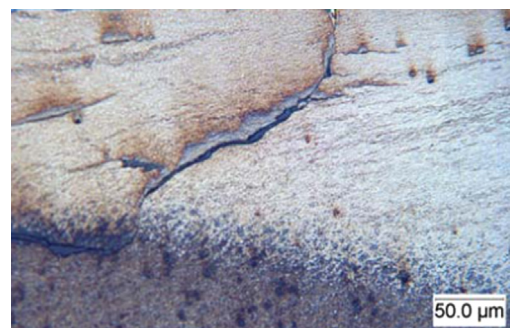


Fig. 6. Wind turbine gearbox bearing WECs [31].



Fig. 7. Fatigue fracture of blade bearing inner raceway [47].

Table 1

Wear phenomenon of main wear types.

Wear type	Wear phenomenon
Adhesive wear	Material transfer, plastic deformation [48]
Abrasive wear	Scratches, grooves [48]
Corrosion	Red/brown stains or deposits [34]
Brinelling	Indentations [49,50]

catastrophic failure of large-scale wind turbine bearings [24,30,34,33,32].

b) **Lubricant contamination:** Lubricant contamination is another major cause of large-scale wind turbine bearing failures which can dent the bearing raceways. Wear debris, residual particles, dirty lubricants and water are the most common sources of contaminations [35,36]. Fig. 4 shows wear debris particles embedded in the gearbox bearing roller surface.

3.3. Electrical arc erosion

Electric arcing or arc discharge is an electric current breakdown at the connective surfaces between rolling elements and raceway [37]. Electric arc can remove materials from the contact surface [23]. There are possibilities for electric arcing to contain excessive voltage (electrical pitting) and current leakage (electrical fluting); this commonly happens to wind turbine generator bearings [37,38].

a) **Excessive voltage/electrical pitting:** Excessive voltage (electrical pitting) is intentionally or accidentally induced by build-

ing up particles or charged lubricants. When voltage differences are induced between the bearing house and the shaft because of the dissymmetrical effect, the current subsequently produce sparks and arcing, which could melt the bearing's surface [39].

b) **Current leakage/electrical fluting:** Current leakage (electrical fluting) occurs from a relatively lower continuous AC or DC current [29,38], so it is gentler compared with electrical pitting. In the beginning, the damaged surface resembles shallow craters which are closely located. After that, clusters of shallow craters form the equally spaced flutes in a dark color, which can only be found on rollers and raceway surfaces [23]. Fluting is the washboard effect on raceways which causes noise and vibration of the damaged surface [29]. Fig. 5 shows a schema of a bearing electrical fluting.

3.4. Cracks and fractures

Cracks are the materials failures and fractures are the results of crack propagation. A number of factors, such as temperature, stress concentration, thermal shock and speed of loading, can affect the fractures of materials [40]. Cracks and fractures are the predominant failure modes for large-scale wind turbine bearings, which can lead to significant repair costs. The followings are some specific failure modes of cracks and fractures:

a) **Forced fracture:** If the stress concentration of the material tensile strength is excessive, the bearing can suffer a forced fracture [23]. White etch cracks (WECs) are a form of forced fracture which may be created by overloaded tensile stresses [41–43]. The appearances of WECs are metallographically similar to butterflies, which can be seen in Fig. 6. The connections of small WECs may form larger WEC networks or white structure flaking (WSF) leading to severe noise and vibration [44].

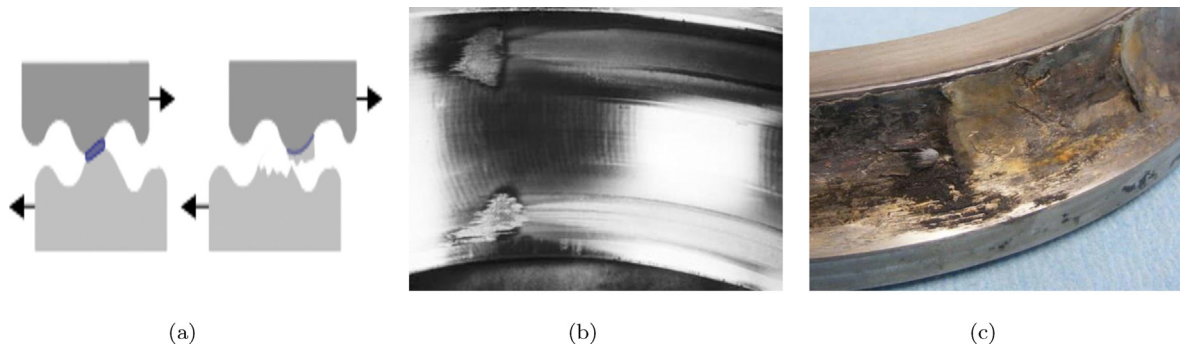


Fig. 8. (a) Mechanisms of adhesive wear, (b) smearing of bearing outer ring [57] and (c) galling of bearing outer ring [57].

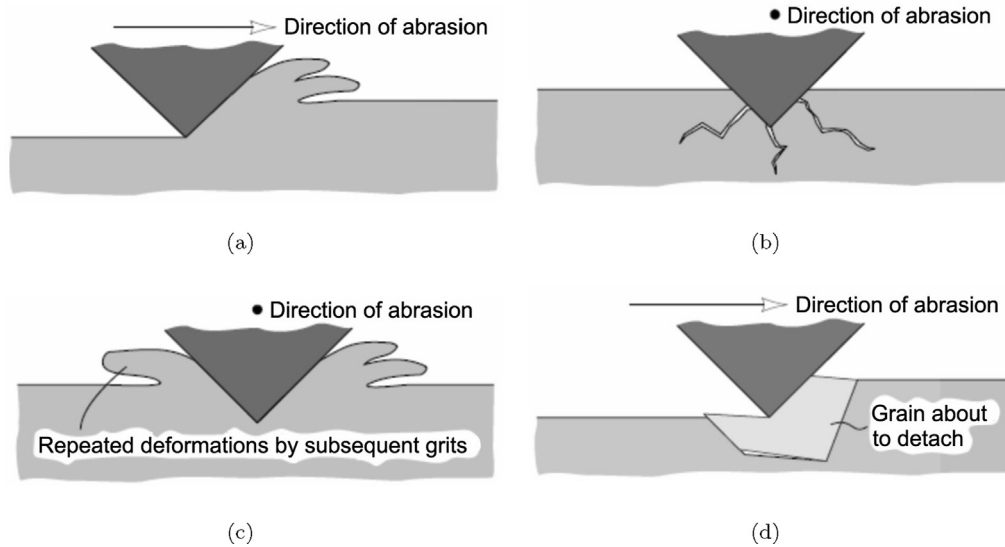


Fig. 9. Mechanisms of abrasive wear: (a) microcutting, (b) microfracture, (c) pull-out and (d) individual grains [40].



Fig. 10. Abrasive wear of wind turbine gearbox bearing inner raceway with visible dents [31].

b) **Fatigue fracture:** Fatigue fracture, also called fatigue failure or spalling (Fig. 7), are fractures to the running surface [45,34]. If the limitations of fatigue strength are frequently exceeded

through bending, tension or torsion conditions, fatigue fractures can occur [23].

c) **Thermal cracking:** High frictional heating due to a sliding motion can lead to thermal cracking. Cracks always vertically happen to the bearing movement direction in relation to the contacting surfaces [46]. As a result, misalignments, lubricant failures, contaminations and bearing overheating are the incentives that may cause thermal cracking [46].

3.5. Wear

Wear is another type of material failure, which happens when the base body and counterbody come into contact with poor lubricant [48]. Wear is defined as a continual loss of material by the mechanical action of a solid, fluid or gaseous counterbody [40,48]. The removed materials are wear particles and debris which can change the bearing frictional characteristics. According to the type of tribological load and removed materials, there are four types of wear process: adhesive wear, abrasive wear, brinelling and corrosion. Table. 1 gives the typical wear phenomena of these four wear types.

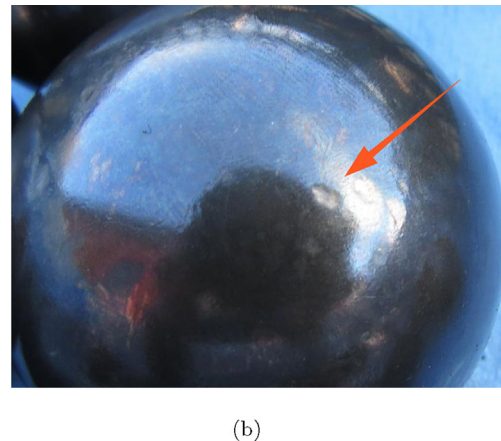
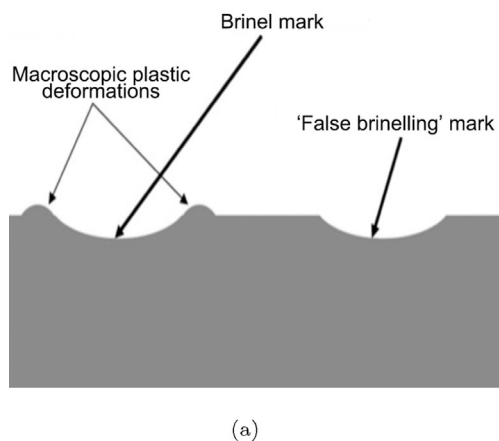


Fig. 11. (a) The schema of true brinelling (left) and false brinelling (right) [49,50] with (b) false brinelling on a blade bearing roller surface [47].



Fig. 12. Corrosion phenomenon on a wind turbine gearbox bearing inner raceway [58].

a) **Adhesive wear:** Adhesive wear happens due to relative motion between contacting surfaces, resulting in plastic deformation and material transfer from one surface to another [23,29,51], which can be seen in Fig. 8(a). Therefore, adhesive wear modifies the original surface texture [29]. According to different damage levels, adhesive wear is defined by different names: scuffing, seizing, smearing, skidding and galling. Scuffing indicates the least severe situation. Seizing, smearing and skidding show it middle severe degree (see Fig. 8(b)), while galling represents the most serious and extensive plastic flow and metal transfer [23,29], as shown in Fig. 8(c).

b) **Abrasive wear/three body wear:** Abrasive wear is also called particle wear or three body wear [23]. The most intensive abrasive wear happens when a soft surface is cut by another hard surface. The second reason occurs when a third hard particle abrades another soft surface [23]. The mechanisms for abrasive wear include microcutting, microfracture, pull-out and individual grains, which are displayed in Fig. 9 [40,52]. Therefore, lubricant failures and contamination are incentives to cause wind turbine bearings abrasive wear [53–55]. This can be seen in Fig. 10. Adhesive wear and abrasive wear are two common failure modes for wind turbine transmission system bearings [56].

c) **True brinelling/False brinelling:** True brinelling happens when loads are over the elastic limit given to the bearing materials. Indentations in the raceways are called as brinell marks, and can lead to vibration noise. False brinelling is similar to true brinelling, but has no macroscopic plastic deformation. Fig. 11 (a) shows the schema for true brinelling (left) and false brinelling (right) [49,50], while Fig. 11(b) displays false brinelling on a blade bearing roller surface.

d) **Corrosion:** Reference [23] defines that corrosion is a chemical reaction to bearing surfaces (Fig. 12). Corrosion mainly contains two types which are moisture corrosion and frictional corrosion. The moisture corrosion happens when a bearing is in contact with moisture (water or acid), while frictional corrosion is activated by relative movements between mating surfaces given certain friction conditions [23].

3.6. Mounting failure/improper mounting

Improper mounting can cause the imbalance load, improper load distributions, uneven heating and misalignments of bearings [24]. Before installation, it is necessary to confirm that the size of the shaft and housing are ready for the bearing [59]. A tiny error will lead to unpredictable vibration levels and excessive temperatures. The followings are two possible results of mounting failure:

a) **Bearing overheating:** Overheating is one of the results of mounting failure generated within the bearing itself which

happens on rings, balls and cages. The symptoms are discoloration from gold to blue [46]. Fig. 13 gives an example of the overheating blade bearing outer ring surface [47]. The incentives for overheating contain heavy loads, inadequate heat paths and poor cooling systems. As a result, both transmission system bearings and adjustment system bearings can easily be subjected to this kind of failure. The inappropriate treatment of overheating has the potential to damage or even explode the wind turbine.

b) **Inadequate bearing selection:** The possibility of inadequate bearing failure should be taken into consideration after other types of failures have been researched [60,24]. The most common mistake is choosing a larger or stronger bearing improperly which can easily lead to overloading and vibration.

3.7. Failure mode classification

There are many causes and modes for bearing failures which have been introduced in the previous subsections. In general, one failure is induced via many reasons and there is a combination of causes. During the last decades, the development of tribological research has led to a noticeable increase in outstanding failure mechanisms and failure modes classifications. For example, ISO 15243:2004 [23] classified six groups of failure modes based on bearing failure types. For this paper, based on the classifications in Refs. [23,40], the failure modes of large-scale bearings introduced in the previous subsections are classified into two main aspects: failure of material and premature fatigue (see Fig. 14). The failure of materials is caused by components and structures, and can only be reflected with cracks, wear and other similar phenomena [40]. The second aspect is premature fatigue which is based on mechanical failures. All bearing failures are attributed to either material failures or premature fatigue. The majority of topics on premature fatigue are on plastic deformation, improper lubricant, electrical erosion, improper mounting and improper maintenance.

Compared with ISO 15243:2004 [23], Fig. 14 provides a more detailed summary and each of the specific failure modes for wind turbine bearings can be easily categorized. However, the faults from wind turbines are not isolated and this type of classification cannot reflect the interrelationship between premature fatigue and material failures. Fig. 15 gives four networks illustrating the causes and effects of bearing failure and shows a comprehensive bearing failure analysis.

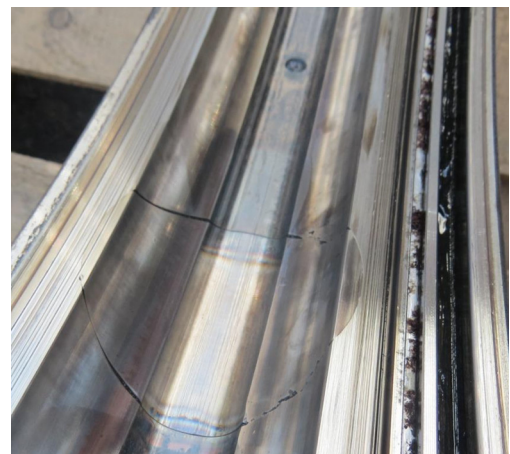


Fig. 13. Overheating blade bearing outer ring surface [47].

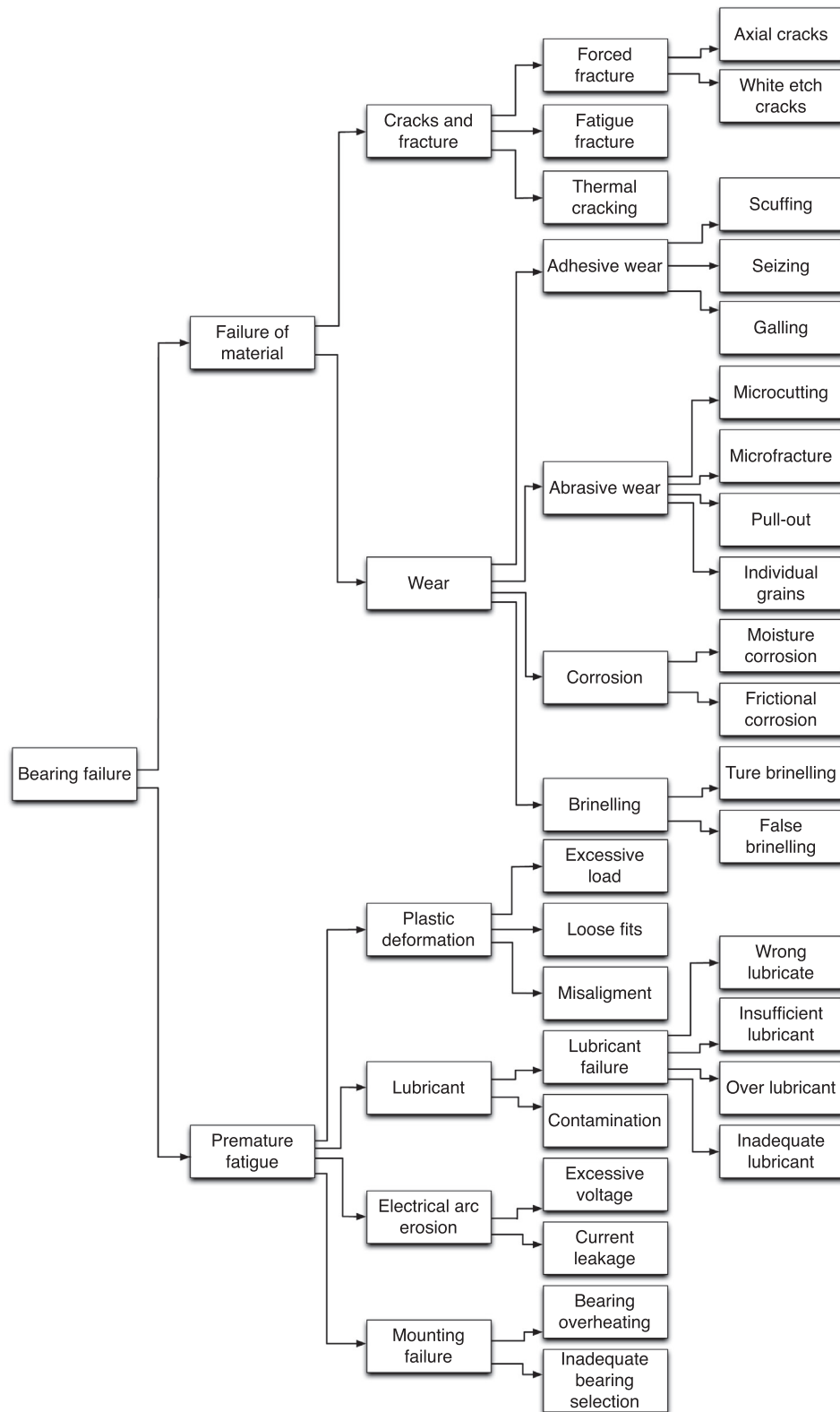


Fig. 14. Failure mode classifications of large-scale wind turbine bearings [23,40].

After having classified bearing failure modes, it is necessary to conclude the specific fault types of large-scale wind turbine bearings. As mentioned before, large-scale bearings working under different working conditions, such as environment, temperature and load distribution, may generate some particular failures. For the

fault diagnosis of wind turbine bearings, if researchers can master these failure modes, it may be easier to identify the particular bearing fault types combining some signal processing analyses and bearing working conditions. Table 2 concludes the most common failure modes of specific large-scale wind turbine bearings.

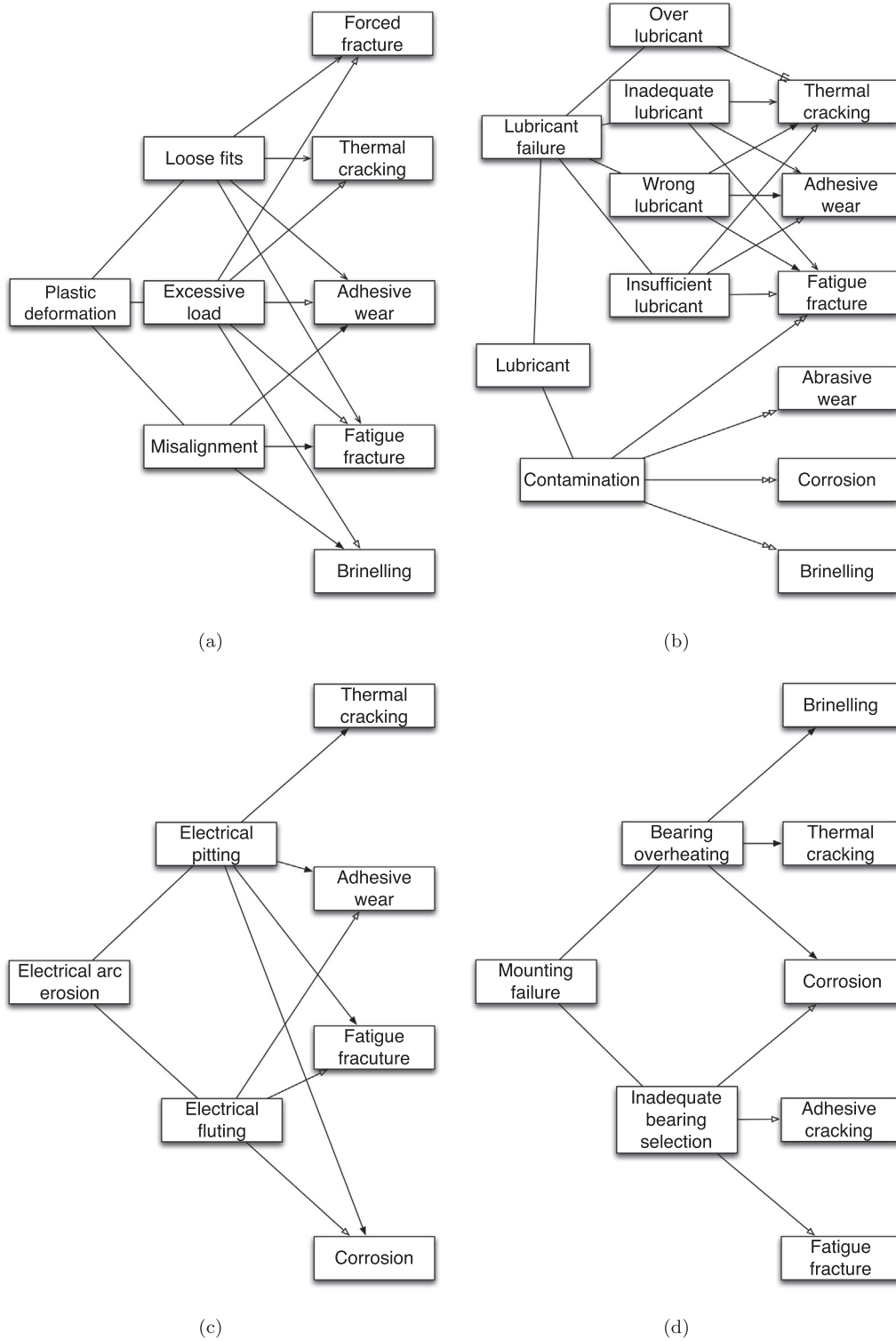


Fig. 15. Interrelationships between (a) plastic deformation (b) lubricant (c) electrical arc (d) mounting failure/improper mounting and materials failures [23,40].

4. Review of CMFD methods for large-scale wind turbine bearings

The previous section classified the failure modes of large-scale wind turbine bearings. In order to evaluate bearing health conditions, it is important to develop CMFD systems which can detect, identify, localize and even prognose the failures of wind turbine bearings [61]. In this section, both existing and potential CMFD methods for large-scale wind turbine bearings are summarized.

4.1. Data acquisition techniques

Data acquisition techniques have been developed to collect failure information using different sensors or equipment. These techniques have been validated for large-scale wind turbine bearings CMFD including vibration analysis, acoustic emission (AE) analysis, lubricant/debris analysis, power quality analysis and microscope analysis, which are introduced in the following subsections.

Table 2
Failure modes of large-scale wind turbine bearings [23,40].

			Transmission bearings			Adjustment bearings	
			Generator bearing	Main bearing	Gearbox bearing	Blade bearing	Yaw bearing
Material failure	Cracks and fracture	Forced fracture	X	X	X	X	X
		Fatigue fracture	X	X	X	X	X
		Thermal cracking	X	X	X		
	Wear	Adhesive wear	X	X	X		
		Abrasive wear	X	X	X		
		Corrosion				X	X
		Brinelling				X	X
Premature fatigue	Plastic deformation	Excessive load	X	X	X	X	X
		Loose fits	X	X	X		
		misalignment	X	X	X		
	Electrical arc erosion	Excessive voltage	X				
		Current leakage	X				
	Mounting failure	Overheating	X	X	X	X	X
		Inadequate bearing					
	Lubricant	Lubricant failure	X	X	X	X	X
		Contamination	X	X	X	X	X

4.1.1. Vibration

Vibration due to a fault happens when a ball strikes the fault in the track. Vibration signals can be collected using vibration sensors including accelerometers, velocity sensors and displacement sensors; accelerometers are the most commonly used sensors [61]. The construction of an accelerometer contains several piezoelectric-crystal elements and a mass. The main characteristic of accelerometers is that they have a wide frequency range. At present, vibration analysis is the most popular and useful measurement for wind turbine bearing condition monitoring, due to its wide frequency range (often 1–30 kHz) and easy usage.

4.1.2. AE

The publication of [62] reports that if the system has fast pressure pulses with a period of around 5 us, the normal accelerometers may not react accurately because their natural frequencies are not high enough. Although a piezoresistive accelerometer usually has wide bandwidth, from a few hundred Hz to over 130 kHz, this kind of accelerometer generally has low sensitivity [63].

An AE transducer works in the frequency range from 20 kHz to 1 MHz [64]. Fundamental to AE is the phenomenon of radiation of elastic wave generation in solids under stress. If the solid undergoes stress over a particular level, the fast release of strain energy will be transmitted in the form of elastic waves which can be collected by the AE sensor [65]. At present, the AE method is already used in small-scale bearing condition monitoring, but there are very few publications on the diagnosis of large-scale wind turbine bearings.

4.1.3. Lubricant/debris

Machinery lubricant systems always include an amount of solid debris which are from surroundings (minerals), component manufacture (core sand, weld spatter, grinding debris) and machinery operation (wear debris). The lubricating oils and grease of large-scale wind turbine bearings can be analysed by using filters, magnetic plugs, spectrographic oil analysis, ferrographic oil analysis, and X-ray fluorescence [66]. Tough or large hard debris potentially initiates rolling contact fatigues, leading to severe surface damage. In order to solve this issue, filtration systems are designed to remove large debris, thus increasing the bearing life cycles. Nonetheless, small debris which is greater than the lubricant film thickness still damages the bearing surface as it is too small to remove. The materials on the bearing surface will be removed after repeated damage by such small debris [67].

4.1.4. Power quality

In the future, wind farms will be required to meet the increasing criteria of grid integration of wind energy, so the power quality of wind energy will become more reliable than before [68]. The faults of mechanical components, such as bearings, can be detected by analysing the power quality of mechanical or electrical energy measured on a shaft or generator [8]. Peak power output, reactive power, voltage fluctuations, current fluctuations and harmonics are the main factors indicating power quality [69]. Also, the Supervisory Control and Data Acquisition (SCADA) system is another method to provide detailed power quality measurements that indicate whether bearings need to be maintained or replaced [70].

4.1.5. Microscope

Microscope analysis is to visually understand the microstructure or nanostructure of failure modes, such as WECs, erosion and wear debris. The images of the failure modes can be captured using a variety of methods, such as X-ray microscopy, scanning electron microscopes (SEM), transmission electron microscopes (TEM), confocal and laser scanning microscopes. After the images are captured, they can be stored, analysed and displayed via a computer [71]. Moreover, based on the images, the bearing fault properties can be deduced.

4.1.6. Other techniques

There are many other data acquisition techniques, such as shock pulse meter (SPM), temperature monitoring, sound analysis and jerk measurement, which have been applied to normal bearings although very few publications have focused on large-scale wind turbine bearings. For future works, these techniques may need researchers to explore on the large-scale wind turbine bearings in detail.

4.2. CMFD for main bearings

4.2.1. Vibration signal analysis

Wind turbine main bearings work under non-stationary conditions with dynamic loadings, so the inconstant vibration signals are commonly transformed into stationary signals or decomposed into several quasi-stationary components with the aim of extracting weak fault vibration features. The current achievements of vibration analysis when used for main bearings basically contain time-frequency-energy analysis (e.g. Ref. [72]), frequency domain analysis (e.g. Refs. [73–75]), time-frequency analysis (e.g. Refs. [76,77]) and artificial intelligence (AI) methods (e.g. Ref. [78]).

An et al. [72] used the intrinsic time-scale decomposition (ITD) method to decompose the vibration signal of the non-stationary direct-drive wind turbine main bearing into several proper rotation components. For this, both instantaneous amplitude and frequency are defined. The dynamic features used for fault diagnosis were extracted from the proper rotation components. A lab scale experiment was performed to identify the artificial inner race crack, outer race crack and roller crack of the bearings.

Xin et al. [73] and Pan et al. [74] conducted a similar experiment. They received vibration signals of two main bearings from two 1.5 MW direct-drive wind turbines. One bearing worked normally and the other had a fault. The different characteristics of a normal bearing and a faulty bearing were compared by using cepstrum analysis [73], narrow-band envelope analysis [73] and spectral correlation analysis [74]. However, the authors were unable to diagnose the exact bearing fault type.

Zhang et al. [75] firstly proposed the generalized transmissibility damage indicator (GTDI) method which was based on transmissibility function (TF) analysis. This can be defined as the ratio of spectral of two different output measurements. The advantage of GTDI is that it can eliminate dynamic loadings and noise effects. After that, the authors conducted a field test to diagnose wind turbine main bearings via the proposed method. The results showed that GTDI has the potential for online fault diagnosis because the bearing fault can be rapidly detected (around 1.5 s) without misrecognition. Based on this technique, Zhang et al. [79] developed a novel wavelet energy transmissibility function (WETF) which can further reduce the effects of non-stationary operations and noise. The results of the analysis can better evaluate the health conditions of wind turbine main bearings.

Bodla et al. [76] proposed Hilbert-Huang transform (HHT), feature extraction and logistic regression (LR) for a real data set of 18 wind turbines to evaluate the health condition of the wind turbine main bearings with an accuracy of 87%. Chen et al. [77] proposed a multifractal method based on the wavelet transform modulus maxima (WTMM) method to investigate the non-stationary vibration signals of wind turbine main bearings. The field test was conducted and the data was collected from a real wind farm. The results showed that incipient fault features can be extracted from a multifractal spectrum. Han et al. [80] utilized Teager energy operator for signal enhancement and Complementary Ensemble Empirical Mode Decomposition (CEEMD) for signal denoising. A scale-down wind turbine was built, and a seeded crack was induced in the outer race of the main bearing with the aim of validating the proposed diagnostic framework. The results showed that CCEMD could effectively filter the background noise and obtain bearing defect frequency components.

Purajomandlangrudi et al. [78] combined Anomaly Detection (AD) and Support Vector Machine (SVM) machine learning algorithms for wind turbine main bearing fault detection. The research

conducted a lab scale experiment proving that the AD method performs better than SVM to detect incipient bearing faults.

4.2.2. Stator current analysis

For direct-drive wind turbines, the main bearing health conditions can be detected via generator stator current signals; this is because the main shaft is directly connected to the generator and any faults in the main bearings will pollute the generator current signals. The researches of references [81–86] analysed stator current signals to diagnose the main bearings of direct-drive wind turbines.

Gong et al. [81] applied a wavelet filter to stator current signals which can increase the signal-to-noise ratio (SNR) of the raw data and extract incipient faults. Gong et al. [82–84] used stator current power spectral density (PSD) analysis on stator current signals. The method converted the variable fundamental frequency of the stator current into a fixed frequency in order to identify the bearing faults. Wang et al. [85] proposed the multiscale filtering spectrum (MFS) method to transform the time domain signal into energy distribution in the order domain based on the Vold-Kalman filter (VKF) and the bearing fault characteristic orders can be identified from the weighted energy spectrum. Wang et al. [86] applied a current-aided order tracking method to directly extract the shaft rotating frequency from the stator current signal. Based on the extracted shaft rotation frequency, the raw variable speed vibration data can be resampled into a constant vibration signal which can then be easily used for bearing fault diagnosis.

The references of [82–86] performed lab and field scale experiments to evaluate the proposed methods which can effectively detect naturally damaged and artificially damaged main bearings. The failure modes of the bearings are outer ring fracture, inner ring fracture and cage fracture.

4.2.3. Summary of the review for main bearings CMFD

The signals used for wind turbine main bearings mainly contain vibration and stator current signals. Table 3 provides a comprehensive summary of using these signals for main bearings CMFD. As can be seen, the CMFD techniques for main bearings are merely implemented at the experimental level. Seeded faults can be diagnosed under stationary and non-stationary conditions at relatively high rotation speeds. However, for the field test, the current techniques can only present a fault alarm (e.g. with a distinct fault or without a distinct fault).

4.3. CMFD for gearbox bearings

4.3.1. Vibration signal analysis

A wind turbine gearbox has three stages and each stage produces different bearing rotation speeds. This complicated design heavily increases the difficulty of collecting vibration fault signals

Table 3
Summary of the review for main bearings CMFD.

Signals	Ref(s).	Processing methods	Problem(s)	Experiment	Failure modes
Vibration	[72]	ITD	Non-stationary	Lab scale	Inner, outer, roller cracks
	[73]	Cepstrum/band-pass analysis	Characteristic comparison	Field test	Fault alarm
	[74]	Spectral Correlation analysis	Characteristic comparison	Field test	Fault alarm
	[75]	GTDI	Non-stationary/noise	Field test	Fault alarm
	[79]	WETF	Non-stationary/noise	Field test	Fault alarm
	[76]	FFT, HHT and LR	Non-stationary/noise	Field test	Fault alarm
	[77]	WTMM	Non-stationary	Field test	Incipient fault extraction
	[80]	CEEMD	Noise	Field test	Seeded outer race crack
	[78]	AD/SVM	Noise	Lab scale	Incipient fault extraction
Current	[81]	Wavelet filter	Noise	Lab scale	incipient fault extraction
	[82–84]	PSD	Non-stationary	Lab scale	Inner, outer, cage cracks
	[85]	MFS	Non-stationary	Lab scale	Inner, outer, cage cracks
	[86]	Current-aided order tracking	Non-stationary	Lab scale	Inner, outer, cage cracks

because the useful information can be submerged by noise. The signal processing methods of [87] utilized time domain analysis. Refs. [88–93] used frequency domain analysis and Refs. [94–99] used time-frequency analysis. These can be explained as follows.

Wang [87] proposed the autoregressive (AR) filter for wind turbine gearbox bearing vibration signals. The gearbox noise was eliminated via the AR filter and the fault signals were retained in the residual signals. After that, the envelope method was applied to the filtered signals in order to find the bearing fault frequencies. The data for the research was provided by a lab scale experiment and the bearing inner race spalling was identified.

Xu et al. [91] collected vibration signals of a naturally damaged wind turbine gearbox and its fault type was unknown. After analysing the raw data in both time domain and frequency domain, the authors found that the planetary bearing most likely had faults. Finally, endoscope inspection and disassembly of the gearbox confirmed that the vibration signal analysis was accurate. Antoniadou et al. [89] performed a lab scale experiment to diagnose seeded inner race electrical erosion with 0.2 mm deep and 0.2 mm wide via envelope analysis. Sawalhi et al. [90] presented both the envelope method and cepstrum method to diagnose different stages of wind turbine gearbox bearings. The research data was provided by National Renewable Energy Laboratory (NREL). The results showed that the high-speed stage bearing with overheating, and a minor scratch on the inner raceway and balls were detected; while, the intermediate stage bearings with debris, false brinelling and corrosion were not diagnosed. Jain et al. [92] combined the high-pass filtering and envelope method to process the vibration data of a wind turbine planetary gearbox bearing. The proposed method is able to diagnose the artificial fracture of a bearing's inner raceway and outer raceway. Shanbr et al. [93] combined a fast kurtogram with the envelope method which detected inner race faults on high-speed stage gearbox bearings. Peeters et al. [88] utilized the cepstral editing procedure analysis to separate gears and shaft noise from raw vibration data. After that, envelope spectrum analysis was applied to the denoised signal with the aim of finding the bearing defect frequencies. The data of this research was provided by the NREL. The diagnostic results showed that the faults on intermediate-speed stage bearings and high-speed stage bearings could be confidently identified. However, the researchers were not confident with the diagnosis results of low-speed stage bearings as the envelope spectrum spike is too small to make any conclusion.

The researches of [94,95] used undecimated wavelet transform and second-generation wavelet transform to decompose wind turbine gearbox bearing vibration signals. Then, a proper analysis, such as frequency analysis or envelope analysis, was applied to the decomposed components in order to identify the bearing fault types. Ref. [94] identified the inner race crack of a high speed gearbox bearing from a field test and Ref. [95] identified the bearing inner race fault from a lab test.

Hu et al. [96] combined ensemble empirical mode decomposition (EEMD) and SVM to diagnose the noise masked non-stationary vibration signals of a wind turbine gearbox bearing. Li et al. [97] combined variational mode decomposition (VMD) and convolutive blind-source separation (CBSS) to process vibration signals of multi-channels. The first stage of [96,97] was to decompose the raw vibration data into a collection of intrinsic mode functions (IMFs). Compared with empirical mode decomposition (EMD), EEMD and VMD have the ability to eliminate the mode mixing problem. Finally, Hu et al. [96] identified an inner race fault, outer race fault and rolling element fault via SVM, while Li et al. [97] diagnosed artificial cracks on bearing outer raceway where the cracks had three different depths (i.e., 1.0, 2.0, 3.3 mm). Additionally, Guo et al. [98] further proposed the local mean decomposition (LMD) to diagnose the high-speed stage gearbox bearing. The

LMD was firstly proposed by Smith to decompose amplitude and frequency modulated signals into a small set of product functions (PFs) [99]. Each PF is defined as the product of an envelope signal and a frequency modulated signal. From PF, a time-varying instantaneous phase and instantaneous frequency can be collected. Similar to the EMD-based approach, LMD can also separate the noise in decomposed PFs. Thereafter, synchrosqueezing transform was applied to convert the defined PF in the time-frequency plane where fault frequencies could be seen. This proposed method successfully diagnosed bearing inner race scratch under non-stationary conditions. Zhang et al. [100] combined VMD and adaptive maximum correlated kurtosis deconvolution (AMCKD) to filter raw vibration signals. The authors conducted different cases studies to validate the proposed diagnostic framework where the damages on inner race and balls were diagnosed.

Teng et al. [101] used artificial neural networks (ANN) to train data-driven models in order to predict the remaining useful life of the wind turbine gearbox bearing. The researchers conducted a field test which collected the vibration signals of 1.5 MW wind turbine gearbox bearings. The prognosis results can predict several days before the failure happens. Saari et al. [102] utilized SVM for the bearing fault identification. The diagnostic results showed that the wind turbine gearbox bearing inner race spalling was detected. Ali et al. [103] proposed adaptive resonance theory 2 (ART2) for an unsupervised classification which could classify the degradation of wind turbine gearbox bearing into three status (failure, degraded and healthy). Compared with conventional feature extraction methods, such as time-domain analysis, frequency domain analysis and time-frequency analysis, the proposed method can estimate the bearing degradation level immediately. Li et al. [104] developed a reliability assessment method, degradation-Hidden-Markov model (DHMM), which can provide the guide for wind farms to make optimal operation and maintenance strategies. The advantage of DHMM is that it can achieve the reliability analysis with fewer sampling data when compared with traditional reliability analysis methods, such as Hidden-Markov model (HMM). The author verified the proposed method on a 1.5 MW wind turbine gearbox bearing and found that the bearing began to degradation from the 99th month.

4.3.2. Electrical signal analysis

Zhang et al. [105] used electrical signature analysis (ESA) for monitoring gearbox bearings in wind turbines with the aim of avoiding mechanical noise. The signal processing method of ESA in [105] was the electrical multiphase imbalance separation technique (eMIST) which was able to improve the signal-to-noise-ratio (SNR) of the raw electrical signals by transforming the healthy signals to DC components and measuring the difference between the healthy condition and the faulty conditions of the bearings. After that, bearing fault types can be identified in the eMIST spectrum.

The proposed algorithm was validated by a lab test and a field test. For the lab test, the vibration signals were received from a 25-hp wind drivetrain simulator. The experimental results indicated that the proposed algorithm was capable of diagnosing a high speed bearing artificial inner race crack and outer race crack (1–2 mm deep and 1 mm wide). In the field test, the authors collected electrical signals for 1.5 MW double-fed induction generators (DFIG). The proposed method can diagnose inner race cracks of high-speed stage bearings and outer race cracks of planetary bearings.

4.3.3. Microscope analysis

Evans et al. [44] designed a large-scale industrial transient test rig to simulate the real working conditions of wind turbine gearbox bearings. Transient speeds with dynamic loads were applied to the

gearbox bearing which is able to reproduce WECs and WSF on the bearing inner ring. After analysing the WECs and WSF via optical microscopy, the authors deduced that the formation of WECs and WSF relates to the steel cleanliness because MnS associated inclusions (D_{sulf} and D_{Dup}). Duplicates and type D inclusions were the incentive to form WECs and WSF.

Bruce et al. [106] used optical microscopy and electron microscopy to observe the WECs on wind turbine gearbox bearings. The authors found that the WECs are subsurface-initiated by manganese sulphide (MnS) inclusions.

Gould et al. [107] used X-ray tomography to determine the characteristics of wind turbine gearbox bearing WECs. After observing four crack networks on the bearing inner ring, three properties of WECs were concluded. Firstly, WECs can be formed in the bearing without direct connection with the bearing raceway. That is to say, lubricant is not a necessary factor in causing WECs. Secondly, an improper running condition is an element for accelerating the formation of WECs. Lastly, some inclusions in WECs lead to the direction of crack growth.

Gegner et al. [108] applied X-ray diffraction based residual stress analysis to a wind turbine gearbox bearing. The author concluded that the fatigue cracks were induced by brittle spontaneous incipient cracks from the bearing raceway surface.

4.3.4. Lubricant/debris analysis

Dupuis [109] utilized oil debris monitoring sensors to diagnose the health condition of a wind turbine gearbox bearing. Large wear debris over 200 μm proved that the bearing surfaces had the fault of spalling. Moreover, the rate of generation for particles could be used as an indicator to evaluate the remaining useful life of the bearing.

4.3.5. Summary of the review for gearbox bearings CMFD

For the CMFD of wind turbine gearbox bearings, signals including vibration, electrical, microscope and lubricant are analysed. Table 4 summarizes the survey of gearbox bearing CMFD methods. For vibration and electrical signal analysis, most researches (e.g. Refs. [87,101,94–97,88–90,100,102,104,103,93,98]) have investigated fault detection for gearbox bearings on intermediate or high-speed stages. However, the CMFD for low-speed stage

bearings (e.g. planetary bearing) is at an initial phase as the rotation speeds are very low (10–15 rpm) and the potential fault signals are extremely weak. The only available publications are Refs. [88,91,92]. With regard to microscope and lubricant analysis, the properties of WECs, fatigue cracks and spalling are investigated.

4.4. CMFD for generator bearings

4.4.1. Vibration signal analysis

For vibration analysis of generator bearings, the bearing rotation speeds are higher than any other large-scale wind turbine bearings and they may vary heavily during operation. Therefore, a primary difficulty in CMFD of wind turbine generator bearings is to extract fault features from non-stationary and noise masked raw signals. To solve these problems, some researchers (e.g. [110–112]) utilized stochastic resonance (SR) based methods and other scholars (e.g. Refs. [113–118]) used the wavelet transform, EMD, and their variational forms, to process the inconstant signals and minimize noise levels. Additionally, scholars [118–120] used the order tracking technique to resample a non-stationary vibration signal into a stationary or quasi-stationary signal in the order domain (angular domain) which was helpful for further fault diagnosis analysis.

In order to denoise raw signal and extract weak fault features, SR-based methods are developed for wind turbine generator bearing fault diagnosis. The first step of SR is the frequency shift modulation which shifts the original signal with the carrier frequency f_0 to the desired frequency $f_0 + f_1$ and $f_0 - f_1$. Then, some parameter tuning SR methods, such as quantum particle swarm optimization (OPSO) [110] and discrete wavelet transform (DWT) [112], are used to detect the weak fault signals. In this framework, Li et al. [111] developed adaptive SR method based on coupled bistable systems (ACBSR) to further improve the SR filtering performance, which had a great ability to suppress low-frequency and high-frequency interferences. The proposed diagnostic frameworks of Refs. [112,111] were verified on a high-speed wind turbine generator bearing. The diagnostic results showed that the proposed framework could clearly demonstrate bearing inner race defect frequency.

Table 4
Summary of the survey for gearbox bearings CMFD.

Signals	Ref.	Processing methods	Problem(s)	Experiment	Speed	Failure modes
Vibration	[87]	AR filter	Noise	Lab scale	High	Inner race spalling
	[91]	frequency analysis	Unknown conditions	Lab scale	Low	Fault alarm
	[89]	Envelope analysis	Noise	Lab scale	High	Seeded inner race erosion
	[90]	Envelope/Cepstrum	Noise	Simulation	High	Inner overheating/scratch
	[92]	High-pass/envelope	Noise	Lab scale	Low	Seeded Inner race cracks
	[93]	Kurtogram/envelope	Noise	Lab scale	High	Inner race fault
	[88]	Cepstral editing	Noise	Simulation	High	Inner,outer,roller fault
	[94]	Wavelet-based	Noise	Field test	High	Inner race crack
	[95]	Wavelet-based	Noise	Lab scale	High	Inner race fault
	[96]	EEMD/SVM	Non-stationary/noise	Lab scale	High	Inner,outer,roller fault
	[97]	VMD/CBSS	Non-stationary/noise	Lab scale	High	Seeded outer race cracks
	[98]	LMD	Non-stationary/noise	Lab scale	High	Inner race scratch
	[100]	VMD and AMCKD	Noise	Lab scale	High	Inner and ball fault
	[101]	ANN	Fault prediction	Field test	High	Fault prognosis
	[102]	One-class SVM	False alarms	Field test	High	Inner race spalling
	[104]	DHMM	Reliability assessment method	Field test	High	Bearing Degradation
	[103]	ART2	Non-stationary/noise	Lab scale	High	Fault classification
Electrical	[105]	eMIST	Noise	Lab/Field	High	Naturally damaged cracks
Microscope	[44]	Optical microscopy	Failure mode properties	Lab scale		WECs and WSF
	[106]	Optical microscopy	Failure mode properties	Lab scale		WECs
	[107]	X-ray tomography	Failure mode properties	Lab scale		WECs
	[108]	X-ray diffraction	Failure mode properties	Lab scale		Fatigue cracks
Lubricant	[109]	Oil debris monitoring	Unknown condition	Lab scale		Spalling

Zimroz et al. [121] utilized the linear regression analysis for the longtime statistical data to implement and interpret the working condition of wind turbine generator bearing. The research triggered two case studies. The first presented the development of a wind turbine generator bearing fault. The second tracked data from a damaged bearing and its replaced bearing. Some differences can be seen before and after the damaged bearing replaced. Teng et al. [115] conducted a large scale field test for a 2 MW wind turbine generator bearing. The vibration signals were analysed via the cyclic coherence function, conventional demodulation analysis, complex wavelet transform and spectral kurtosis methods. However, only the cyclic coherence function could detect the bearing inner race fault using intensive electromagnetic vibration (EV). The defects were proven after disassembling the bearing. The authors predicted the defective inner race was caused by shaft current corrosion. Liu et al. [119] used both the cyclostationary demodulation method and envelope method to extract defect shock impulses from the converted angular domain signal. The experiment of [119] was a lab test and the inner race fault of direct-driven wind turbine generator bearing was identified.

Sun et al. [113] used the multiwavelet denoising method with the data-driven block threshold technique to extract the defect shock spikes and eliminate noise. The research firstly performed a simulation experiment to prove the efficiency of the proposed method. After that, a field test for a real wind turbine generator bearing was conducted which detected an incipient inner race fault. Cheng et al. [116] proposed the empirical wavelet transform (EWT), combining the advantages of wavelet transform and EMD to extract inherent modulation information by decomposing a noise masked non-stationary signal into mono-components under an orthogonal basis. The research performed three experiments to evaluate the proposed method. The first one was a lab scale experiment which diagnosed the bearing outer ring with one crack. The second experiment was also a lab scale test for compound fault detection which identified outer race spalling and roller scrape. The last experiment was a field test for a real wind turbine generator bearing. Spalling in the outer race was detected. Teng et al. [122] utilized parameterless-based EWT to automatically segment raw vibration signal in the frequency domain. The segmentation approaches were based on the local maxima method and scale-space method that are capable to deal with the massive data. The authors diagnosed a 1.5 MW wind turbine generator bearing where the ball fault, inner race fault and cage fault were diagnosed. He et al. [114] proposed ensemble super-wavelet transform (ESW) to investigate vibration characteristics of wind turbine generator bearings. The ESW was a combination of wavelet transform and Hilbert transform, which could effectively extract weak fault vibration features. The authors applied a field test for a wind turbine generator front bearing and back bearing. The proposed method could diagnose fault types with a slight scrape defect on the inner race of the front bearing.

Song et al. [118] transformed a non-stationary vibration signal into a stationary angular domain signal and detected impulse in the angular domain by using EMD and Hilbert transform. The generator bearing outer race groove was detected after a lab scale experiment. Mollasalehi et al. [123] applied a field test to acquire vibration signals from a wind turbine generator bearing. The raw data was decomposed via the EEMD method and an outer race fault was diagnosed. Zheng et al. [117] used VMD, a methodology for the adaptive and quasi-orthogonal signal decomposition, which is based on Wiener filtering and frequency mixing. The signal could be decomposed into various modes or IMFs to extract the incipient fault feature. The experiment was a lab scale which concluded an outer race fault in the bearing. Chen et al. [124] implemented VMD for raw signal denoising as it is insensitive to noise and extracts fewer IMFs when compared with EMD. The VMD denois-

ing was verified by two experiments including a simulation and a lab test. The vibration signals of the lab tests were provided by the Case Western Reserve University (CWRU) Bearing Data Center, which were acquired from normal bearings, inner race fault bearings, outer race fault bearings and ball fault bearings. After filtering the raw signals, SVM was utilized in order to classify the bearing fault types. The diagnostic results showed that the combination of VMD and SVM could identify the bearing fault type with the accuracy of 98.75% which was supervisor to other EMD-based techniques.

Pezzani et al. [120] proposed a novel order tracking technique to resample the non-stationary vibration signal into a stationary signal under the estimated rotor position. This rotor position was obtained from the angle of the voltage vector which was calculated by using the phase-locked loop (PLL) method. Finally, FFT was applied to the resampled signals in order to identify the specific bearing fault type. To validate the proposed method, the authors conducted a lab scale experiment to collect a 3.5 kW permanent magnet synchronous machine (PMSM) electrical signals and faulty bearing vibration signals. The results showed that the proposed method can identify different sizes of artificial fractures on a bearing outer raceway. The diameter of the fractures are 2 mm, 4 mm and 6 mm. Guo et al. [125] utilized the generator current signal as a reference to resample the non-stationary vibration signal into the stationary signal. As the resampled signal submerged noise interference, an adaptive stochastic resonance (ASR) filter is used to denoise the signal as well as enhance the weak fault signals. To verify the utilized methods, the authors conducted four experiments to diagnose the seeded defect bearings which have different sizes and types of faults.

Liu et al. [126] applied the Integral Extension Local Mean Decomposition (IELMD) method to extract obvious features from non-stationary signals. The experiment was a lab test which detected the generator bearing inner race fracture and outer race fracture. Chen et al. [127] applied the Nonlocal means algorithm (NLmeans) to extract the fault features and minimize the background noise in the time-domain. This approach discovered the inherent repeatability of the fault signals and tried to recover the original fault signals using the average of other similar ones. To demonstrate the feasibility of the NLmeans, the authors applied several simulations and experimental case studies. The results indicated that the generator bearing outer race crack and pitting fault can be diagnosed in both the frequency domain and the order domain.

Yang et al. [128] developed a data-driven fault diagnosis method based on the sparse representation and the shift-invariant dictionary learning for wind turbine generator bearings, which extracted different impulsive components from vibration signals. The advantage of this method is that it can select a proper dictionary without being dependent on prior knowledge. The research performed two experiments. The first research was a lab scale experiment which used an artificial outer ring scratched bearing to show that the proposed method has the ability to distinguish the bearing health condition. The second experiment was a field test for a real wind turbine generator bearing, which diagnosed that the fault was on the bearing inner race. Sun et al. [129] utilized the sparse representation technique and dictionary learning to denoise the raw signal and extract weak fault signals. The author implemented a field test to diagnose a 2 MW wind turbine generator bearing. The diagnostic results indicated that the bearing most likely had an inner race fault. Guo et al. [130] used the k-means singular value decomposition (K-SVD) technique as the dictionary learning method. The field test was conducted to validate the proposed method where the bearing outer race spalling fault was diagnosed. Wang et al. [131,132] improved a new sparse representation method for gen-

Table 5

Summary of the review for generator bearings CMFD.

Signals	Ref(s).	Processing methods	Problem(s)	Experiment	Failure modes
Vibration	[121]	Linear regression	Fault prediction	Lab scale	Fault alarm
	[110]	SR	Noise	Field scale	inner race fault
	[112]	SR	Noise	Lab scale	inner race fault
	[111]	ACBSR	Noise	Field scale	inner race fault
	[115]	Cyclic coherence function	Noise	Field scale	Seeded inner race corrosion
	[119]	Cyclostationary	Non-stationary/noise	Lab scale	inner race fault
	[113]	Multiwavelet denoising	Noise	Field test	Inner race fault
	[116]	EWT	Non-stationary/noise	Lab/Field	Crack, spalling and scrape
	[122]	Parameterless EWT	noise	Field	Inner, ball and cage fault
	[114]	ESW	Noise	Field test	Inner race scrape
	[118]	EMD/order tracking method	Non-stationary/noise	Lab scale	Outer race groove
	[123]	EEMD	Noise	Field test	Outer race fault
	[117]	VMD	Noise	Lab scale	Outer race fault
	[124]	VMD	Noise	Lab scale	Inner, outer and balls
	[120]	FFT and PLL	Non-stationary/noise	Lab scale	Seeded outer race fracture
	[125]	ASR	Non-stationary/noise	Lab scale	Seeded inner and outer race fault
	[126]	IELMD	Non-stationary	Lab scale	Seeded inner, outer race fault
	[127]	NLmeans	Non-stationary/noise	Lab scale	outer race crack/pitting
	[128]	Spare representation	Non-stationary/noise	Lab/field	Inner, outer scratch/fault
	[129]	Spare representation	noise	field	Inner race fault
	[130]	Spare representation	noise	field	Outer race spalling
	[131]	Spare representation	Non-stationary/noise	Lab/field	Inner, outer fault
Electrical	[133]	Electrical signature analysis	Non-stationary	Lab scale	Seeded outer race defect
	[134–139]	EMD, EEMD and HHT	Non-stationary/noise	Lab scale	Inner, outer, cage fracture
	[140]	MSB	Non-stationary/noise	Field	Electrical corrosion
SCADA	[141]	Neural networks	Fault prediction	Field test	Fault prognosis

erator bearings under non-stationary conditions. For the new spare representation method, a new dictionary consisting of a time-varying cosine packet has been designed which is able to vary with the shaft rotation frequency. The solved non-zero sparse coefficients related to the bearing fault types. The proposed method was validated by a lab test and a field test. From the lab test, the seeded outer race fault was identified; and from the field test, the naturally damaged generator bearing inner race fault was diagnosed. As a result, the spare representation can diagnose the bearing fault types under stationary and non-stationary conditions, but an improper dictionary choice will limit the success of the sparse representation method.

4.4.2. Electrical signal analysis

The faulty bearing can induce the wind turbine shaft fluctuation and further result in the modulations of the stator current at the bearing defect frequencies. In this scenario, the electrical signal collected from the wind turbine generator can be analysed to diagnose the bearing fault. For example, Shahriar et al. [133] acquired the generator current signal to estimate the bearing rotation speed and resample the non-stationary current signal; therefore, the bearing fault type can be diagnosed in the order domain based on the resampled signal. Amirat et al. [134–139] proposed EMD, EEMD and HHT to diagnose bearing fault types by using the wind turbine generator stator current signal. The signal was decomposed into a number of IMFs, with each of these a mono-component. After locating the most energized mode, the envelope method or statistical criterion analysis was applied to IMFs in order to reveal the generator bearing fault. The authors presented a lab based experiment which used artificial failures of generator bearings to prove that the proposed method has the ability to diagnose bearing fault types. The failures of the bearings were outer race fracture, inner race fracture, cage deterioration and ball deterioration. Chen et al. [140] utilized the modulation signal bispectrum (MSB) for the generator bearing fault diagnosis. The advantage of MSB is that it can avert the misdiagnosis by including phase information of stator current. The diagnostic framework was tested on a

wind farm with thirty-three 1.5 MW wind turbines. The failures subjected to electrical corrosion were identified.

4.4.3. Supervisory control and data acquisition (SCADA) analysis

Kusiak et al. [141] used neural network algorithms to build fault prediction models for wind turbine generator bearings. The data of bearing's temperature was collected at twenty-four 1.5 MW wind turbines over a period of four months. The result of the research predicted 1.5 h before the occurrence of any bearing faults.

4.4.4. Summary of the review for generator bearings CMFD

Table 5 concludes the survey of generator bearings vibration CMFD approaches. For vibration and current signal analysis, the proposed methods can extract noise masked raw data and diagnose the bearing faults under constant and variable speed conditions. However, only very few publications can be found for SCADA analysis which could be considered as a research direction for wind turbine generator bearings CMFD.

4.5. CMFD for blade bearings and yaw bearings

Wind turbine blade bearings and yaw bearings failure leads to poor pitching, low energy output and turbine shutdown. However, to our best knowledge, there is no standard method and very few publications are available for blade/yaw bearing CMFD. The only reference about blade bearing fault detection is that Liu et al. [142] developed empirical wavelet thresholding method which is the combination of empirical wavelet transform and wavelet thresholding to diagnose a naturally damaged wind turbine blade bearing. Furthermore, the references about fatigue life analysis are summarized as follows. The methods include frequency spectrum analysis [143], circular domain analysis [144], spectrum kurtosis [145], peak-peak value analysis [145], average value analysis [146] and static load-carrying capacity analysis [147], which can detect changes in the condition of the blade/yaw bearing and track the bearing progressive deterioration during the long-time of measurement days by comparing working conditions of the good bear-

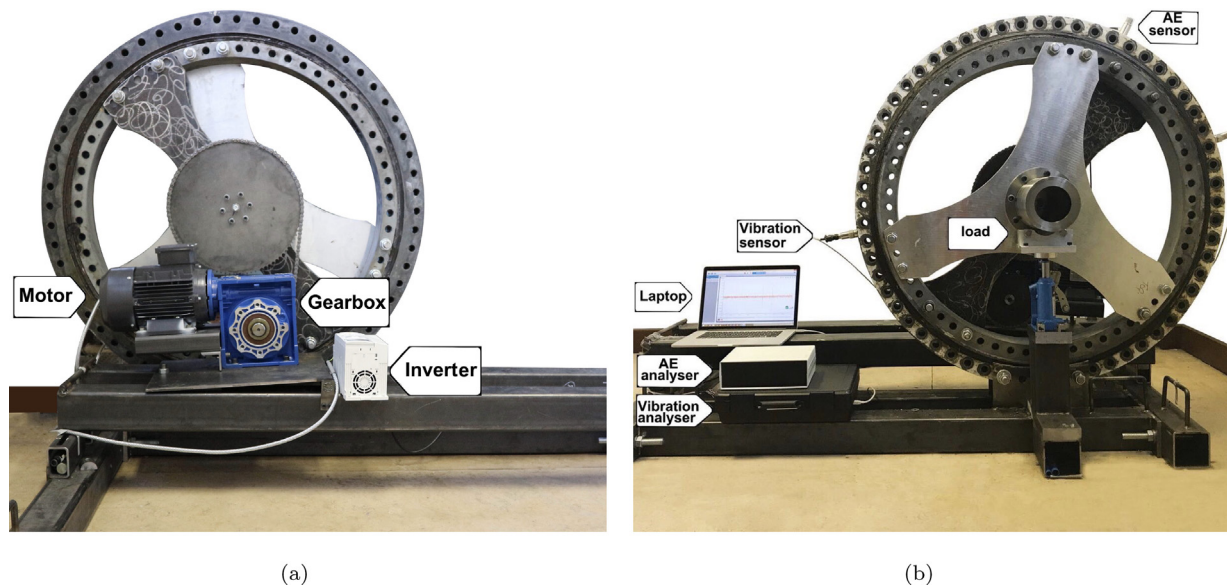


Fig. 16. The blade bearing test rig (a) front view (b) back view.

ing. However, in practice, the working conditions of good blade/yaw bearing are usually not recorded for spare use. Therefore, there is a gap in blade/yaw bearing condition monitoring as the above-mentioned methods used in the fatigue life analysis may not be used in real cases.

In order to fill the gap for diagnosing adjustment systems bearings, the authors designed a blade bearing lab at the University of Manchester, as shown in Fig. 16. The outer ring of the bearing is fixed on the test rig, so the inner ring can be rotated for fault diagnosis. As can be seen in Fig. 16(a), the kinematic components of the test rig include the three-phase induction motor, gearbox and the blade bearing. The motor has the ability to drive the bearing in the test. Due to the gearbox, the maximum bearing rotation speed is reduced to 5 rpm. The motor inverter can adjust bearing rotation speeds from 0.5 rpm to 5 rpm. Fig. 16(b) displays the back view of the test rig. The bearing load can be adjusted via a jack or the bearing can be assembled with a real wind turbine blade for a load in order to simulate wind turbine blade bearing working conditions. The vibration and AE signals are acquired from the accelerometers and AE sensors which are mounted on the bearing outer ring surface; and the vibration analyser and AE analyser are used to power the sensor, strip off the bias voltage and output signals. Finally, the data is collected using a laptop. This rig can ideally imitate the field situations because real blade bearings in wind turbines rotate slowly with heavy loads [142].

4.6. Potential CMFD methods for large-scale wind turbine bearings

The previous subsections have comprehensively reviewed existing works on large-scale wind turbine bearings CMFD. However, the technique of wind energy is still in an initial stage due to its short history. Some popular CMFD methods have been used in other areas, especially in terms of the slow-speed, non-stationary and artificial intelligent (AI) aspects, which can potentially be used for large-scale wind turbine bearings.

With respect to slow-speed operations, it seems that AE signal analysis and lubricant analysis can diagnose bearing faults at an early stage. In respect of non-stationary operations, the research trend for bearings in CMFD aims at improving non-linear, non-stationary feature extraction and fault classification algorithms to enhance fault diagnosis sensitivity and selectivity [148]. As for AI

methods, they have attracted great attention from numerous fields and industries, including finance, education, transportations and rotating machines, but their practicality still needs to be verified on large-scale wind turbine bearings.

4.6.1. Slow-speed sensing

a) **AE:** Recently, there has been a great increase in the research of AE analysis on high-speed rolling element bearings, whereas, the analysis of AE has not been very fruitful on slow-speed bearings, let alone large-scale and slow-speed wind turbine bearings. The main difficulties are as follows. Firstly, the AE signal is not strong and sensitive enough under very low rotation speed conditions. At a very low speed, the AE signal is primarily generated by the friction and rubbing between the rollers and raceway, while at a high speed, the AE signal is generated from the strike between two metal surfaces [149]. Furthermore, the sampling rates of the AE signal are much higher than the traditional vibration technique, normally from 500Ks/s to 40Ms/s. As the rotation speeds are very slow, in order to collect enough useful defect signals, the sampling time needs to take long enough resulting in making data processing difficult. To address these issues, the publications on CMFD of slow-speed bearings via AE signal analysis have been reviewed. Caesarendra et al. [149] reviewed some related articles and their signal processing approaches are mainly of conventional time domain and frequency domain analysis. In the last three years, in order to process a large amount data, some researchers have utilized EMD-based, EEMD-based and wavelet-based time-frequency domain analysis to decompose signals into different time scales, and then some data reduction approaches can be applied, such as Independent Component Analysis (ICA) and Principal Component Analysis (PCA), in order to recover the decomposed AE signals with higher signal-to-noise ratio. For example, Nikolakopoulos et al. [150] applied EMD and the envelope method to diagnose a slewing bearing with an inner race fault. Kang et al. [151] proposed the wavelet package transform to detect slow-speed bearings with different sizes of artificial inner race faults, outer race faults and ball faults. Aye et al. [152] utilized a PCA-based algorithm to successfully detect a bearing fault deterioration under varying slow speed condi-

tions. Žvokelj et al. [153] combined EEMD and ICA to diagnose an accelerated run-to-failure slewing bearing with rotation speeds of 1 to 8 rpm. This proposed method can successfully diagnose the bearing cracks and spall on an outer ring surface.

b) **Lubricant:** Monitoring and analysing the contaminations of lubrication plays an important role as it can diagnose the tribological change of wind turbine bearings at an early stage. However, only one related publication [109] can be found for wind turbine gearbox bearings. References are drawn from other publications on slow speed bearings lubricant analysis, as summarized below. Bai et al. [154] analysed the health condition of ship loader slewing bearings and stacking crane slewing bearings via oil analysis. The authors suggested that the contents of Fe and Cr in oil samples were higher than the standard value which indicates the test bearings were worn and extrinsic contaminant particles could be removed from the lubrication by adding extra solvent. Patel et al. [155] applied a lab test to analyse the effect of different liquid contaminants, such as water, diesel, and salt water, in lubrication on the bearing failure. The results show that a high contamination concentration level in lubrication can lead to abnormal shake in the bearing and an increase in temperature.

4.6.2. Non-stationary methods

a) **Instantaneous angular speed (IAS):** The principle of using an IAS signal as the CMFD tool is that the defects in the rotation machine have an instantaneous impact on the angular speed of the rotatory shaft where the IAS sensors, e.g. optical encoder and hall effect tooth sensors, can be used to detect variations [156]. Therefore, IAS with its great noise suppression performance has attracted significant attentions regarding rotating machines. For example, IAS-based approaches have been developed and verified on high-speed rotors, gearboxes, diesel engines, bearings and motors [157–160]. Li et al. [158] utilized EMD to decompose the IAS signal into several IMFs and applied Auto-correlation Local Cepstrum (ALC) for the fault detection of a multistage gearbox. Zhou et al. [159] proposed EEMD to decompose the IAS signal and applied the envelope method to the defined IMF to diagnose a feed-axis gearbox. However, the drawback of the IAS method is that the IAS signal is easily disturbed by random load fluctuations, shaft misalignment and structure noise. In order to fill this gap, Li et al. [160] developed an instantaneous angular phase demodulation (IAPD) method which gathers normalization, narrowband filter and Hilbert phase demodulation to pre-process the raw IAS signal with the aim of increasing the IAS estimation accuracy.

b) **Improved order tracking:** For the CMFD of wind turbines, order tracking has been applied to main bearings and generator bearings which is an effective approach for minimizing the effect of nonstationary speeds. However, this method requires an additional tachometer or encoder to record rotation speeds which will increase the cost. Additionally, the installation of a tachometer may not be convenient for some bearings. As a result, traditional order tracking methods still have some challenges for large-scale wind turbine bearing fault diagnosis. To address these issues, Lu et al. [161] improved a fast and online order analysis (FOOA) to realize nonstationary permanent magnet synchronous motor (PMSM) bearing fault detection. The authors utilized a current sensor to record the rotating phase information from the PMSM sinusoidal current and a microphone to collect the bearing sound signal with the aim of exposing the bearing fault characteristic order for fault identification. Wang et al. [162] proposed a computer-vision-based rotation speed estimation method for motor bearing diagnosis. Firstly, a high-speed camera was used to record and analyse the instan-

aneous rotating speed (IRS) of the motor. Meanwhile, a microphone was provided for order tracking based on the obtained IRS curve. These improved order tracking methods avoid the installation of a tachometer; however, for wind turbines, the additional instruments also increase measurement costs.

c) **Tachless order tracking:** In practice, due to the technical and economical limitations, some instrumentation based order tracking methods may hardly be satisfied for large-scale wind turbine bearing fault detection. A further improvement is the tachless order tracking approach which can extract the speed reference signal from the raw vibration data itself. From Ref. [163], time-frequency representation (TFR) is seen as a powerful and reliable approach for non-stationary signal analysis. Extracting time-frequency curves from the TFR can be applied to estimate the instantaneous rotating speed. Refs. [164,165] introduced band-pass filtering methods to extract the instantaneous rotating speed from the raw vibration data where the chosen frequency band contains shaft speed variability. Hu et al. [166] presented a dynamic path optimization based ridge detection (DPORD) to extract the time-frequency curves which can overcome the drawbacks of noise interference and the local optimal. Huang et al. [167] proposed multiple time-frequency curve extraction from the TFR, based on a fast path optimization which is more reliable than single ridge curve extraction. Whereas, the proposed methods above are only applicable with restricted speed fluctuations. For large speed variation conditions, the instantaneous frequency ridges of different components on the time-frequency plane demonstrate spectrum overlap. Wang et al. [163] proposed a tachless filtering method with Vold-Kalman filtering (VKF) which can separate mono-components from time-varying signals for instantaneous phase extraction. This method is capable of working on larger speed variation conditions. As a result, tachless order tracking methods can reduce measurement costs and effectively diagnose variable speed rotation machines; however, these approaches are usually complicated and require massive computation costs.

d) **Stochastic resonance (SR):** SR is a non-linear phenomenon where a weak signal can be enhanced by adding an optimal amount of noise, as the frequencies of the added noise and the weak signal will resonate with each other to make the weak signal prominent [168]. Therefore, SR is a potential tool for diagnosing large-scale wind turbine bearings with incipient faults. Guo et al. [169] utilized the SR method to filter the vibration signal collected from a permanent magnet synchronous generator (PMSG) bearing. The results show that the proposed method can enhance the weak fault features and facilitate bearing fault identification. Lei et al. [170] employed the adaptive resonance stochastic (ASR) to extract weak fault signatures of a planetary gearbox. The method of ASR is to optimize bistable SR parameters via ant colony algorithms. The ASR incorporated SR weak signal enhancement and overcame untoward SR parameter selections. Qiao et al. [171] developed an adaptive unsaturated bistable SR (AUBSR) to avoid the output saturation of SR. This method has verified on high speed bearings with slight flaking faults and a planetary gearbox with a chipped tooth and a missing tooth.

These methods proposed above have effectively been proved on high-speed rotation machines, but their sensitivity, validity and feasibility still need to be verified on wind turbine bearings with slow variation speeds and large dynamic loadings.

4.6.3. AI methods

AI methods have achieved great success on traditional rotating machines CMFD; and Liu et al. [172] and Cerrada et al. [173] have reviewed the AI methods applied to fault severity assessments in

traditional rotating machines where the widely used AI methods contain statistical and classifiers learning approaches including SVM, ANN, k-nearest neighbor (k-NN) algorithms, Bayesian classifier, Markov model-based methods and even deep learning approaches [174,175]. However, AI methods are still in the prototype stage for wind turbine bearing CMFD because there are only a few AI methods applied to the transmission system bearings (main bearings, gearbox bearings and generator bearings). Some selected state-of-the-art AI methods which have been successfully applied to traditional rotating machines are briefly reviewed as they can be exploited for large-scale wind turbine bearings CMFD.

The machine fault diagnosis follows a procedure of data acquisition, feature extraction and fault classification [176]. As the raw signals always have redundancy information, such as noise disturbances, it is very difficult to directly utilize the collected raw signals for fault classification. In consequence, feature extraction is a vital process which can extract the diagnostic information for future fault classification. One of the popular AI methods for feature extraction is the sparse representation and dictionary learning which can sparsely describe the raw signals based on the designed dictionary; therefore, the redundancy information is eliminated and the fault signals are extracted. In this diagnostic framework, Yang et al. [177] proposed a data-driven fault diagnosis method based on sparse representation and shift-invariant dictionary learning, which represented the impulse signals at different locations with the same characteristic by only one atom through shift operation. The effectiveness of this method was verified by two experiments. Zhao et al. [178] developed a dictionary learning method where the dictionary was built from a union of segmented raw signals. This process simplified the dictionary development. Huang et al. [179] proposed a new dictionary learning method, adapted dictionary free orthogonal matching pursuit (ADOMP) for incipient fault features extraction. Based on sparse representation theory and Wigner-Ville distribution, Yang et al. [180] proposed a sparse time-frequency representation method for incipient fault diagnosis of wind turbine drive train. Neural networks based methods can directly use the raw signals for fault classification without the process of feature extraction [181]. In this framework, Liu et al. [182] proposed a joint-loss convolutional neural network (CNN) for simultaneous fault diagnosis and remaining useful life prediction. The results of bearing run-to-failure datasets were used to verify the effectiveness of the joint-loss CNN method. Yang et al. [183] proposed a double-convolutional neural network for bearing RUL prediction. The prediction results of four real tests of bearing degradation demonstrated the superiority of the double-CNN method. Liu et al. [184] proposed a dislocated time series convolutional neural architecture based on the characteristics of industrial vibration signals to extract the relationship between signals with different intervals in periodic mechanical signals; therefore it is more applicable for modern electric machines, especially under nonstationary conditions. Zhao et al. [185] proposed deep convolutional neural network (DCNN) for planet bearing fault type recognitions where the classification accuracy exceeded 98.3%. Furthermore, Lei et al. [186] and Fu et al. [187] built a Long Short-term Memory (LSTM) model which is developed from Recurrent neural network (RNN) for wind turbine fault classification. The introduced neural networks based methods can diagnose and recognize the bearing fault types, but they often require large training datasets and the calculations are complicated.

4.7. Summary

In a nutshell, the current CMFD achievements of large-scale wind turbine bearings are reviewed in order to present full-scale information for related researchers. Table 3–5 compare different research

cases based on data acquisition techniques, signal processing methods, main problems, experiment scale and failure modes. From Table 3–5, some experiments can diagnose specific failure modes (e.g. fracture, cracks, and electrical erosion); whereas, for the majority of researches, they only limit identifying bearing fault types (e.g. inner race fault, outer race fault and ball fault) or fault alarms (e.g. with distinct fault and without distinct fault).

With respect to potential CMFD approaches, three main research directions which are slow-speed sensing, non-stationary operations and AI approaches applied to traditional rotation machines are summarized. The knowledge of these applied techniques can be used as a reference for large-scale wind turbine bearings fault diagnosis.

5. Concluding remarks

In conclusion, in order to improve the electric energy output and reduce operation and maintenance costs, it is necessary to monitor large-scale wind turbine bearings. For this paper, it firstly reviewed the types of large-scale wind turbine bearings and their corresponding failure modes, which provides a comprehensive references for researchers and engineers to better understand the fundamental tribological issues and their characteristics and further enable them to propose or utilize more reliable and cost-effective bearing diagnosis methods. After that, the current achievements of large-scale wind turbine bearings condition monitoring and fault diagnosis methods are reviewed. Finally, some promising methods that can deal with the existing problems and future research directions were summarized. In the future, it may be possible to develop research opportunities in the following areas:

- For wind turbine gearboxes bearings, they have three stages i.e. low-speed stage bearings, intermediate-speed stage bearings and high-speed stage bearings. As can be seen in Table 4, the majority of researches still focus on intermediate-speed stage bearings and high-speed stage bearings. However, it is currently in the early stages of development to diagnose low-speed gearbox bearings. A key challenge is that slow rotation speeds result in weak fault signals. The same challenges remain for wind turbine main bearings.
- For wind turbine adjustment system bearings, such as blade bearings and yaw bearings, there are only very limited researches for their condition monitoring and fault diagnosis. Unlike transmission system bearings (main bearings, gearbox bearings and generator bearings), adjustment system bearings are typified as the slow, large and partially rotated slewing bearings which can optimize electric energy output. However, these bearings are the weak part for wind turbines due to their harsh working environment. The failure of adjustment system bearings can easily result in lost control of wind turbines and heavily reduce energy production. Therefore, it is essential to develop reliable and cost-effective condition monitoring and fault diagnosis methods for adjustment system bearings.
- For failure modes assessment of large-scale wind turbine bearings, most of the existing works were only validated with artificially seeded defects in idealized laboratory environments. The designed techniques may not necessarily perform well on field operating wind turbines due to the complex failure modes. Therefore, it is desirable to propose more reliable condition monitoring and fault diagnosis methods for naturally damaged and multiple fault types and to extensively evaluate the diagnosis methods in field tests.
- For data acquisition techniques, the majority condition monitoring and fault diagnosis techniques are based on vibration signal analysis which can only partially assess the bearing

conditions. It is still challenging to detect incipient faults, identify damage severity, determine bearing failure modes and make decisions on repair, replace and shutdown for specific faults. Therefore, there is a need to develop some intelligent methods and apply different combinations of data collection techniques, such as grease, endoscopy and acoustic emission analyses, for large-scale wind turbine bearings with the aim of obtaining more thorough diagnostic results.

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